

RAMALINGAIAH K R

Software Engineer | AI/ML | Generative & Agentic AI

Doddaballapur, Karnataka • +91 7483446584 • ramukr496@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student (B.Tech, 2026) building **Generative AI and agentic systems** end to end. Hands-on with **AI agent development** — designing prompts, instructions, tools, reasoning loops, and guardrails — plus **RAG pipelines, vector databases, embeddings, tokenization, and prompt engineering**. Strong Python engineer who ships **RESTful APIs with FastAPI**, integrates LLM providers and third-party services, and deploys with **Docker and Git/GitHub**. Backed by a six-month full-stack internship and multiple production-style projects across the AI/GenAI lifecycle. Eager to contribute as a Software Engineer / AI/ML / GenAI engineer.

TECHNICAL SKILLS

Generative & Agentic AI: LLM agents (tools, instructions, reasoning, guardrails), RAG pipelines, prompt engineering, tokenization, embeddings, vector databases; LangChain, Hugging Face, OpenAI API, local/open models, Model Context Protocol (MCP); agent frameworks (CrewAI / Autogen / LangGraph)

AI / Machine Learning: Machine learning, deep learning (TensorFlow / Keras / PyTorch), NLP, model training & evaluation

Languages: Python (pandas, NumPy), Java, JavaScript, SQL

Backend & APIs: FastAPI, RESTful API design, third-party/API integration; Spring Boot, Hibernate, Spring Data JPA; microservices principles, cloud computing basics

Data Science: Data cleaning, EDA, statistics, A/B & hypothesis testing; Matplotlib, Seaborn, Looker Studio, Excel

Databases: MySQL, PostgreSQL, SQLite; schema design, query optimization

Tools & Practices: Docker, Git, GitHub (version control), Streamlit, Jupyter, VS Code; Agile/Scrum, code reviews, SDLC, testing

EDUCATION

B.Tech, Computer Science and Engineering 2022 – 2026

Presidency University, Bangalore — CGPA: 7.12

Relevant coursework: Machine Learning, Probability & Statistics, Data Structures, Algorithms, DBMS, Operating Systems.

Class XII (Pre-University), Narayan PU College — 65% 2020 – 2022

Class X (ICSE), Aditya Public School — 88% 2010 – 2020

PROJECTS

PDF Layout-Aware RAG Agent | Python, YOLO11, DocLayNet, Cohere (embed-v4.0 / command-a), Weaviate, PyMuPDF, Streamlit

- Built an end-to-end **RAG pipeline**: trained a YOLO11 layout model on DocLayNet to detect PDF sections and drive layout-aware **chunking**; generated **embeddings** with Cohere embed-v4.0 and stored them in a **Weaviate vector database (cloud)**.
- Engineered a grounded **AI agent** (command-a) with tuned **prompts** that returns answers with section/page citations, minimizing hallucination; delivered as a Streamlit app.

Rosetta — Natural-Language-to-SQL Agent Platform | Python, FastAPI, React, LLM (Groq / Llama 3.3), SQLite, MCP, Docker

- Built a full-stack **agentic application** on a **FastAPI RESTful backend** that translates plain English into SQL, runs it on a live database, introspects schema dynamically, and auto-generates charts.
- Designed a conversational agent with **clarification prompts, reasoning, tool use, a self-healing loop, and defense-in-depth read-only guardrails**; kept the **LLM layer provider-agnostic** (Groq / Gemini / Claude).
- Exposed the database as a **Model Context Protocol (MCP) server** with typed read/write tools; **containerized with Docker** and versioned on GitHub.

AI Chatbot Application | Python, NLP, Deep Learning

- Trained an intent-classification model spanning 50+ intents at 85% accuracy using **tokenization**, lemmatization, and text classification.

NyayaSetu — Full-Stack Web Platform | Java/Django, MySQL, REST, Bootstrap

- Built a real-time platform with role-based dashboards; designed the database schema, implemented features end to end, and collaborated in a team.

EXPERIENCE

Java Full Stack Developer Intern — Tap Academy, Bangalore Jan 2026 – Jun 2026

- Developed, tested, and maintained full-stack Java applications (Spring Boot, MySQL); wrote SQL queries, built **REST APIs**, and worked in **Agile/Scrum with Git/GitHub and code reviews**.